

ENERGY

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THE INVISIBLE DEBT

Energy appears as an economic measure – megawatts, barrels, CO₂ equivalents. Numbers, graphs, reports. But behind these figures lie flows, debts, dominations – and making these flows visible takes more than an economist: an artist, an activist, a cartoonist, a photographer, a painter. Countries growing richer appear to have shrunk their ecological footprint – clean production, green policies, falling emissions. But there is an invisible debt: while consumption continues, dirty production is exported to other parts of the world. They borrow from China, import, and pretend to be clean. This is a mechanism of double hygiene – the image clean, the reality dirty.

"Countries growing richer seem to have lowered their ecological footprints, but in fact they are shifting them onto other countries. They keep on consuming while moving dirty production out."

Since the 1970s Turkey has consumed beyond its biocapacity – that is, it takes more than nature can replenish. This is not a technical detail but an existential reality: we are taking more from the land we live on than it gives us. The ecological footprint is measured in six categories: pasture, carbon, built-up land, agriculture, forest, fish production – each a separate item of debt. When it comes to oil, Turkey has no economically viable reserves – the reserves lie deeper, the cost is high, extraction is not profitable. But because there is no serious turn towards alternative energy sources, external dependence persists, and energy imports are the main source of the current account deficit. The concept of adjusted net savings – national savings that count not only your money but also the loss of natural resources – shows that the growth story is being told by eating into nature's capital.

An economist who grew up in a working-class family in Balıkesir, whose path runs from an economics degree to a master's – in a system where a 2.56 grade point average stands in the way – then to a doctorate, and from Erasmus in Portugal to a research year in Sweden, questions the Environmental Kuznets Curve hypothesis: the assumption that as countries grow, environmental pollution first rises and then falls is wrong. It does not fall, it moves. Ecological flows hidden in international trade are the truth behind the story of clean growth. Research on women's labour meets a similar structural problem: educated women's exit from the labour force cannot be explained by education or culture alone – the systemic lack of care services is decisive. Fossil fuel subsidies stand in the way of climate policy. Everywhere the same pattern: invisible labour, invisible debt, invisible cost.

THE LANGUAGE OF COAL

Climate change is an abstract concept – it does not interest most people. The concept is too big, too distant, too unclear. But when you say "coal," when you frame it as a health risk – air pollution, children's asthma, old people's breathing problems, the smoke rising from the thermal plant's chimney – people respond. One must descend from the abstract to the concrete, from the global concept down to the plant to be built in Aliağa. Starting not from the global concept but from everyday questions confined to the local is the only way to move people to act.

"When it's concrete, people act. Another thermal plant is going to be built in Aliağa, the people here will get sick – that's where it starts. Then you can work your way round to climate change anyway."

A political sociologist born in Istanbul, moving from an economics degree to a sociology doctorate in Canada, researches state-civil society-rent relations. Works in the political science department of a university. Research on

environmental organisations since 2008 — the link between the struggle against coal and health issues, climate change communication, the dynamics of civil society mobilisation. The adoption of the Paris Agreement (2015) became an unexpected source of legitimacy for anti-coal activists — in 2016 this international legal reference provided a foothold for local struggles. The drought in Konya, the withering leaves of olive trees — these say more to people than climate change does. Turkey is an ecologically very rich country — but it also has a high capacity to lay that richness waste. Ecological wealth and the capacity for ecological destruction live in the same body. The developmentalist discourse is one of the most important obstacles facing the struggle — because the promise of growth legitimises destruction.

An activist opposing the Yeşil Yol project goes round the mountains trying to organise the people — but is perceived as an "anarchist." This is the problem of the translation of knowledge: the gap between global theoretical knowledge and local practice cannot be closed with good intentions alone. Starting from the concrete, starting from people's lives — there is no other way. Anti-coal activists have understood this: instead of abstract climate targets, saying "the air in your neighbourhood is being polluted, your child is getting sick" is the language that mobilises people. A bridge can be built from the personal to global theory — but the bridge's footings must stand in the local. In the triangle of state-civil society-rent relations, ecological struggle always remains on the margins — but when it begins locally, the margin can become the centre.

THE CHAIN

Nuclear energy is presented as a solution to climate change — but this is wrong.

"We cannot address nuclear energy on its own. We have to think about it within the nuclear chain. From extracting the uranium raw material to generating electricity, and in the end to plutonium — that stuff at 4,000 dollars a gram, it has the whole world wrapped around its finger."

Radioactive waste, thermal pollution, tsunami risks, storms, earthquakes — the risk list is long and each item is a different face of catastrophe. If sea levels rise, six reactors go under at two degrees and seven more at four — and these are four hundred reactors that must go on being cooled even once they are shut down. At Akkuyu itself, the officials do not know the difference between a tsunami wall and a rising sea. Nuclear energy appears to be a technical problem but it is geopolitical, it is a power relation, it is a matter of sovereignty.

An economist spent two years living in Japan and was in the Hiroshima Peace Park at the time of the 1999 earthquake. The Fukushima disaster (2011) changed their life — they have been to Fukushima three times, turned to nuclear research, begun writing for a news site. Now studying for both a doctorate and a second master's — in sociology and civil society — and coordinating an anti-nuclear platform: a full-time struggle. Sinop, Mersin, İğneada — Turkey's nuclear power plant projects, each a separate risk map. A Black Sea campaign — someone named Hüseyin rows a thousand kilometres over three months, turning the body itself into political action. This is aesthetic-political action in its barest form: body, message, movement.

Nazım Hikmet's ballad of Hiroshima, heard in childhood — the weight carried by a poem has, decades later, turned into a sense of personal responsibility. Chernobyl and Fukushima have turned that ballad into reality. Nuclear energy is marketed as a climate solution, but when the calculation is made along the full nuclear chain — the environmental cost of uranium extraction, the energy consumption of processing, the millennia of radioactivity in the waste — the equation comes out far from clean. The geopolitics of plutonium, international lobbying, show that the energy question is not a technical but a political problem.

LIVING WITHOUT MAKING WASTE

A woman born in the Black Sea region, whose childhood was spent between school and farm life, and whose path ran from assistant teacher in a kindergarten class to studying art at university, has returned to her village — alone. Living alone as a woman in a village, making environmental sensitivity part of everyday life — both a solitude and a

strength. Her very presence gives other women courage; accepting one's shortcomings and taking responsibility creates strength.

"I use art as a tool in my own life and in the region where I live. In order not to serve the same system while criticising it, I first had to practise not producing waste in my own life."

Transforming plastic bags into artistic material, converting waste into expression — music, performance, painting, a multidisciplinary practice. In her village a copper mine pit has been converted into a waste disposal site — she organises a protest exhibition against it, but does not stop at drawing attention: together with others, she creates a space for solutions. A nature, culture and arts association is founded in the village — collective struggle as a civil society organisation is stronger than individual artistic action. The kindergarten children listen better to a story told through art — the most beautiful thing about art is that it touches people. Stepping outside the system is not only an individual choice; being an example, giving others courage, becomes a source of collective power. Even the presence of a woman living alone in a village expands what is possible.

THE WEEKLY RITUAL

A cartoonist who has drawn a weekly column in humour magazines since 2006 — twelve years of unbroken routine. The weekly cartoon is a form of record that combines the political agenda with popular culture and urban observations. When repeated, it ceases to be a column in a humour magazine and becomes a kind of history-telling, a recording project. Almanac, exhibition, calendar formats — the topical evolves into archive, humour into historical document.

Ecology is a field that receives very little attention on the general political agenda. But connections can be made: it is possible to start from current politics and tie it to nuclear energy — the cartoonist's skill lies precisely in these transitions. Work on the "Nuclear Turkish Style" film project began with a director about three years ago — producing visualisations and infographics, the conditions at Akkuyu, the industrial topography of how hydroelectric and nuclear plants are distributed.

The figure of the Standing Man — political action made using the body — shows that when performance, text and visual combine, a completely different communicative power emerges. Just like the message of the rowing body in that Black Sea campaign — movement precedes words. Humour and political message must go together — but this is a very difficult balance. Drawing cartoons about ecology is an effort to carry the agenda's most marginal issue to the centre. Working with other disciplines is essential for sustainability — the cartoonist alone is not enough; one must produce together with the director, the researcher, the activists.

CEREYAN

A photographic artist born in Urfa, who did a master's in New York, has been working for years on a project about water, flow and energy — the Cereyan series. Old photographs collected from second-hand booksellers, worker-family negatives discovered during a workshop at the Silahtarağa Campus — memory is the raw material of photography. The first works were born from these archival finds.

Research on the GAP Project, now running for six or seven years: the dream that Harran would "turn into a sea" — a grandfather's poem — and then the reality. We present nature within a utilitarian framework: we dig out the stones, cut off the water, and celebrate it as development. But the drought that follows becomes one of the causes of war beyond the border. Borders exist but nature has no borders at all — a whirlwind suddenly drives sand and fires into the crops; this image has become the main axis of the project.

At a residency in New York, a drip from a bottle is rigged up to grow beans inside a bed — that is how delicate nature's balance is. Our intervention, on the other hand, is very crude and mistaken. Archaeology, geology, Mesopotamia — from Göbekli Tepe to early Byzantium — places submerged under water, population movements,

saving and circularity. As the geographical scale grows, the boundary between local and global blurs. The Repeat-Cycle Talk Series — from Ali Alper's talk on the cycles of stars to ecological cycles — brings different disciplines together. Cereyan — at once electrical current, water flow and an unexpected event — the word itself is the essence of the project. From cotton fields to ancient cities submerged under water, from population movements to dam construction — each a different moment of the same cycle. Reaching children, seeing that what you do reaches them — this is a knowledge carried in the pocket, one that cannot be put into words but works.

THE MOUNTAIN'S SKELETON

A painter born in Ankara, raised in Istanbul, a fine arts graduate — in the art market since 2004, 10 solo exhibitions. A path beginning in 2004 with industrial spaces and abandoned factories has shaped itself around the theme of waste and absence — abandonment itself is both an aesthetic and a critique. After Gezi in 2013 it moves on to wall graffiti, then to quarries. The marble quarries of Carrara in Italy, various places in Turkey — what you see is the mountain's skeleton: bare, stripped, painful.

"In marble quarries you see the skeleton of the mountain — it really pains me. But I have to find a pictorial language."

The distance between a past landscape and a present rupture is what the canvas is about. The "Fill in the Plant" series: artificial nature, vertical gardens, concreted areas — while millions of trees are being cut, we fool ourselves with a few flowerpots. This artificial green make-up is a selfish solution.

Does a painting's beauty make up for the bad things it shows? A collector will buy this painting, hang it on the wall and see it as a beautiful thing — but does this not erase the message? Perhaps. But the aesthetic choice does not make us forget anything — it creates another way of reminding. The work of art is sold, enters a collection, becomes a prestige object — this contradiction is the question the artist puts to themselves. Does the beautiful object render the critique inoperative? The tension between aestheticising nature and simultaneously criticising it is always felt — but to carry this tension rather than flee from it is a form of honesty.

THE ALTERNATIVE WORD

Today an alternative word has been produced here.

"If quite different people had gathered around energy — bureaucrats, political party members, investors — they would have hammered away at development, hammered away at national sovereignty. But by speaking of cycles, speaking of the domination humans have established over nature, we ended up producing an alternative word on the same subject."

At this table, a different language has been spoken — "nuclear chain" instead of nuclear "energy solution," domination instead of development, cycle instead of growth. And changing the language is changing the gaze.

Documentation is different from being solution-oriented — but it is no less valuable. Just archiving, making things visible and so making a difference to people, may be enough. One can be both activist and artist — there are artists who say "I'm an activist" and artists who don't. What matters is not which of these definitions you choose, but what your work is. Not seeing nature as a separate thing, remembering that we are a part of it — the violence done to nature is the violence done to ourselves. We need to remind people of this; at bottom, we need to love ourselves. Being solution-oriented is not a requirement — sometimes just the presence, the witnessing, the record itself is enough.

Seven people — energy economist, political sociologist, anti-nuclear activist, village-dwelling artist, cartoonist, photographic artist, painter — from different places (Balıkesir, Istanbul, Black Sea, Urfa, Ankara), from different paths, have touched the same question. Each has taken hold of the concept of energy at a different point: economic energy, political energy, nuclear energy, vital energy, communicative energy, transformative energy, aesthetic-ethical energy. Ecological debt calculation, political discourse analysis, the risk map of nuclear disaster, the collective path of artistic

intervention, the weekly record of history, the visualisation of transformation, the questioning of aesthetics — all are different faces of the same question. The slower trees grow, the more energy and heat they give — slowness is accumulation. But everything has a cost. Nuclear, coal, the power station — each takes something from nature and gives it back with some risk attached. This is transformation.

Energy is not an economic measure — it is a cyclical phenomenon, and against it stands the history of human domination. Leaving the language of the system for another language — cycle instead of development, balance instead of sovereignty, saving instead of growth — this is a political act. The concept of socio-economic metabolism — how we organise our surroundings as communities, input from outside, processing inside, output to outside — this framework lifts energy out of the realm of technical problems and turns it into an existential question. And this question, when academia, art and activism come together at an interdisciplinary table, can be asked, multiplying like a rhizome. Memory and history have been excavated at this table — old photographs, weekly cartoons, archaeological remains, Fukushima testimonies. All of them serve one project: "not letting it be forgotten." And not letting things be forgotten is at least as political an act as transforming.